

August 3, 2023

Advanced Calculus Final Exam

NOTE: Show explicitly how you reach your answers. Do not skip lines!

Prob.1 (30) We want to evaluate the double integral

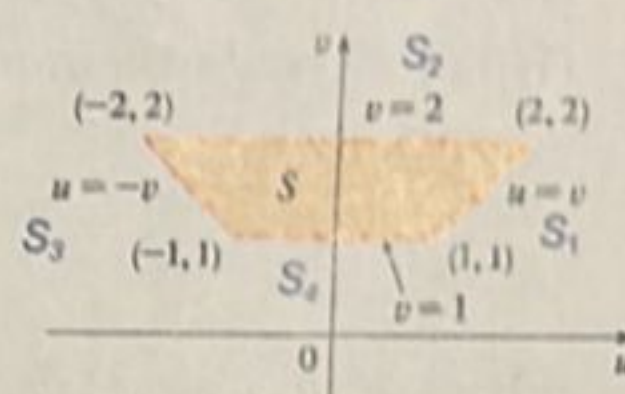
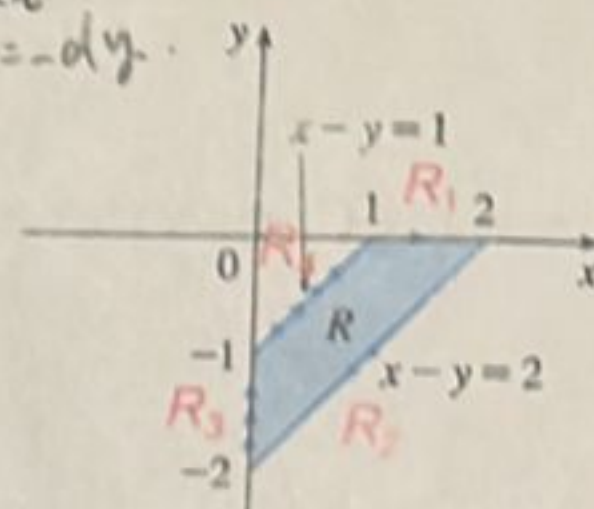
$$I = \iint_R \exp\left(\frac{x+y}{x-y}\right) dx dy,$$

where R is the trapezoidal region with the vertices $(1, 0)$, $(2, 0)$, $(0, -2)$ and $(0, -1)$ (top figure).

(a) Changing the variables by $u = x + y$ and $v = x - y$, show that the region R is transformed to S in the $u-v$ coordinate system (bottom figure).

(b) Find the Jacobian $J = \frac{\partial(x, y)}{\partial(u, v)}$ associated with the change of variables.

(c) Using J in (b), evaluate the double integral I .



Prob.2 (20) The gravitational force $\mathbf{F}$ on a particle with mass m at (x, y, z) due to mass M at the origin is expressed by

$$\mathbf{F}(x, y, z) = -\frac{GMmx}{(x^2 + y^2 + z^2)^{3/2}} \mathbf{i} - \frac{GMmy}{(x^2 + y^2 + z^2)^{3/2}} \mathbf{j} - \frac{GMmz}{(x^2 + y^2 + z^2)^{3/2}} \mathbf{k},$$

where G is the gravitational constant. We want to evaluate the work done by the gravitational force in moving a particle m from point $A(1, 2, 0)$ to $B(2, 4, 0)$ along line C , where the equation of line C is given by

$$C: \frac{x}{1} = \frac{y}{2} = \frac{z}{0}.$$

(a) Suppose that the gravitational potential energy is given by

$$\varphi(x, y, z) = -\frac{GMm}{\sqrt{x^2 + y^2 + z^2}}.$$

Prove that $\mathbf{F}$ is a conservative vector field by showing that $\mathbf{F}(x, y, z) = -\nabla\varphi(x, y, z)$.

(b) Find the parametric equations of line C . In other words, express x, y , and z on C in terms of a parameter t . Then, calculate the work W done by $\mathbf{F}$ by explicitly carrying out the line integral along C , namely, $W = \int_{A \rightarrow B} \mathbf{F}(x, y, z) \cdot d\mathbf{r}$.

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